

Use Case Flows

Smart Expense & Budget Management System

UC-01: Register & Login

Goal	Create an account and sign in securely
Actor	User, with help from the Authentication module
Main flow	1. User selects Sign Up on the landing page. 2. The system displays the registration form. 3. The user enters name, email address and a password. 4. The system checks that the email is not already registered and that the password meets the strength rule. 5. The account is created and the password is stored as a secure hash, never as plain text. 6. The user is taken to the login page. 7. The user enters the email and password and submits. 8. The system invokes Authenticate User to verify the credentials. 9. On success the system issues a session token. 10. The user is taken to the dashboard and the login is complete.
Alternate Flow	Email Already Registered At Step 4, if the email is already in use: 4a. System displays "Email already registered" and does not create the account. 4b. User may sign in with the existing account or use a different email. Invalid Login Credentials At Step 8, if authentication fails: 8a. System displays "Invalid email or password" without saying which one was wrong. 8b. User may retry or start password recovery. 8c. After repeated failed attempts the account is temporarily locked (rate limiting). Session Expired At any later point, if the session has been idle past the timeout: 9a. System ends the session and redirects the user to the login page.

UC-02: Manage Income Records

Goal	Record and maintain income entries
Actor	User

Main flow	1. User selects Add Income. 2. The system displays the income form with today's date filled in. 3. The user enters the amount, the source and the date. 4. The user indicates whether the income is one-time or recurring. 5. The user submits the form. 6. The system validates that the amount is a positive number and the date is not in the future. 7. The income record is saved against the user's account. 8. The system recalculates the available balance for the month. 9. The updated balance and the new entry are displayed. 10. The income record is saved successfully.
Alternate Flow	Editing or Deleting an Entry At Step 1, if the user chooses an existing record instead of adding a new one: 1a. User selects Edit or Delete on an income entry. 1b. On edit, the form opens with the existing values and the flow continues from Step 3. 1c. On delete, the system asks for confirmation and then removes the record. 1d. In both cases the available balance is recalculated straight away. Invalid Income Details At Step 6, if validation fails: 6a. System displays the fields that are wrong, for example "Amount must be greater than zero". 6b. User corrects them and resubmits, or leaves the screen without saving.

UC-03: Manage Expense Records

Goal	Add an expense record
Actor	User, with help from the Analytics Engine and the Notification Service
Main flow	1. User selects Add Expense. 2. The system displays the expense form with today's date filled in. 3. The user enters the amount and a short note or merchant name. 4. The system invokes the Analytics Engine for a category suggestion. 5. The suggested category is displayed and the user confirms or changes it. 6. The user submits the form. 7. The system runs Validate & Categorize Entry to check the amount, date and category. 8. Validation succeeds and the expense is saved against the user's account. 9. The system recalculates the remaining budget for the month and for the category. 10. The system sends the entry to the Analytics Engine for an anomaly check. 11. The updated budget and the new entry are displayed. 12. The expense is recorded successfully.

Alternate Flow	Invalid Expense Details At Step 7, if validation fails: 7a. System displays the fields that are wrong, for example "Amount must be greater than zero". 7b. User corrects them and resubmits, or leaves the screen without saving.
	Budget Exceeded At Step 9, if the expense takes the user past the budget: 9a. System saves the expense, since the entry itself is valid. 9b. A budget alert is shown and the Notification Service is asked to send one. 9c. The remaining budget is displayed as a negative figure.
	Unusual Spending Detected At Step 10, if the amount is well outside the user's normal range: 10a. System flags the entry and asks the user to confirm the amount. 10b. If confirmed, the flag is cleared. If not, the form reopens for editing.

UC-04: Manage Budgets

Goal	Set monthly and category-wise spending limits
Actor	User, with help from the Analytics Engine
Main flow	1. User selects Budgets. 2. The system displays the current budgets along with how much has been spent against each. 3. The user enters a monthly budget for the current month. 4. The user optionally sets category-wise limits for categories such as food or travel. 5. The user may request a recommended budget based on past spending. 6. The system asks the Analytics Engine for suggested limits and displays them for the user to accept or edit. 7. The user submits the budget. 8. The system validates that each limit is a positive number. 9. The budgets are saved and the remaining budget is recalculated. 10. The updated budgets are displayed on the dashboard. 11. The budget is set successfully.
Alternate Flow	Not Enough History for a Recommendation At Step 6, if the account has too little spending history: 6a. System does not produce a recommendation and says that more history is needed. 6b. User enters the limits manually and the flow continues from Step 7.
	Invalid Budget Amount At Step 8, if a limit is zero or negative: 8a. System refuses the value and asks for an amount above zero. 8b. User corrects the figure and resubmits.
	Category Limits Exceed the Monthly Budget At Step 8, if the category limits add up to more than the monthly budget: 8c. System displays a warning showing the total and the difference. 8d. User may adjust the limits, raise the monthly budget, or continue and accept the mismatch.

UC-05: View Dashboard & Analytics

Goal	Understand spending patterns through charts and a forecast
Actor	User, with help from the Analytics / Prediction Engine
Main flow	1. User opens the dashboard after logging in. 2. The system loads the user's income, expenses and budgets for the selected period. 3. The system displays the remaining budget and the total spent so far. 4. The system draws a pie chart of spending by category. 5. The system draws a line chart of spending over time. 6. The system invokes Forecast Monthly Expenditure to predict the expected spend for the month. 7. The forecast is displayed alongside the months it was based on. 8. Any entries flagged as unusual spending are highlighted for review. 9. The user may change the period or filter by category, and the charts refresh. 10. The dashboard is displayed successfully.
Alternate Flow	No Data to Display At Step 2, if the account has no records for the selected period: 2a. System displays an empty state instead of blank charts. 2b. A prompt invites the user to add their first income or expense entry. Not Enough History for a Forecast At Step 6, if there is less than the minimum history required: 6a. System skips the forecast and shows a note that more history is needed. 6b. The rest of the dashboard loads normally and the flow continues from Step 8.

UC-06: Manage Savings Goals

Goal	Create a savings goal and track progress towards it
Actor	User, with help from the Analytics Engine
Main flow	1. User selects Savings Goals. 2. The system displays any existing goals with their current progress. 3. The user selects Add Goal. 4. The user enters a goal name, a target amount and an optional target date. 5. The user submits the goal. 6. The system validates that the target amount is a positive number. 7. The goal is saved and displayed at 0% progress. 8. The system runs Estimate Time to Goal using the recent average saving rate. 9. The estimated number of months is displayed alongside the goal. 10. As income and expenses are recorded, the system updates the progress and the estimate. 11. The savings goal is created and tracked successfully.

Alternate Flow	Invalid Target Amount At Step 6, if the target is zero, negative or not a number: 6a. System refuses the value and asks for a valid target amount. 6b. User corrects it and resubmits. User Is Not Saving Anything At Step 8, if the recent saving rate is zero or negative: 8a. System does not show a number of months. 8b. A message explains that the goal cannot be reached at the current rate. 8c. The system suggests reviewing the budget, and the flow continues from Step 10. Goal Reached At Step 10, if savings reach the target amount: 10a. System marks the goal as completed and shows progress at 100%. 10b. The user is notified and may archive the goal or set a new one.
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UC-07: Manage Recurring Expenses

Goal	Track regular payments and be reminded before they fall due
Actor	User, with help from the Notification Service
Main flow	1. User selects Recurring Expenses. 2. The system displays existing recurring entries with their next due dates. 3. The user selects Add Recurring Expense. 4. The user enters the amount, category, cycle (weekly or monthly) and the start date. 5. The user submits the entry. 6. The system validates the amount, the cycle and the start date. 7. The recurring expense is saved and the next due date is calculated. 8. When a cycle begins, the system creates the expense entry automatically. 9. Before the due date, the system asks the Notification Service to send a reminder. 10. The user marks the payment as paid and the reminder stops for that cycle. 11. The recurring expense is tracked successfully.
Alternate Flow	Invalid Recurring Details At Step 6, if validation fails: 6a. System displays the fields that are wrong and does not save the entry. 6b. User corrects them and resubmits. Duplicate Entry for the Same Cycle At Step 8, if an entry already exists for the current cycle: 8a. System does not create a second entry. 8b. The existing entry is left as it is and the event is written to the log. Payment Not Made At Step 10, if the due date passes without the payment being marked paid: 10a. System marks the payment as overdue. 10b. A follow-up reminder is sent and the entry is highlighted on the dashboard.

UC-08: Run What-If Simulation

Goal	Check whether a planned purchase is affordable
Actor	User, with help from the Analytics / Prediction Engine
Main flow	1. User selects What-If from the dashboard. 2. The system displays the simulator form with the current remaining budget shown for reference. 3. The user enters the planned purchase amount and a category. 4. The user submits the simulation. 5. The system validates the amount and category. 6. The system invokes Forecast Monthly Expenditure to get the expected spending for the month. 7. The system recalculates the forecast with the purchase added, without saving anything. 8. The system works out the remaining budget after the purchase and the effect on any active savings goal. 9. The result is displayed with a verdict of affordable or not affordable. 10. A note confirms that this was a simulation and nothing has been recorded. 11. The simulation is completed.
Alternate Flow	Not Enough History At Step 6, if the account has too little history for a forecast: 6a. System skips the forecast and uses only the remaining budget for the month. 6b. The result is shown with a note that the estimate is based on limited data. Purchase Not Affordable At Step 9, if the purchase takes the user past the budget: 9a. System shows the shortfall amount and the verdict as not affordable. 9b. System suggests an affordable amount, or how many months of saving would be needed. 9c. User may adjust the amount and run the simulation again, or close the result. User Decides to Go Ahead At Step 11, if the user wants to record the purchase for real: 11a. User selects Record this expense on the result screen. 11b. Control passes to UC-03 with the amount and category already filled in. 11c. Nothing is saved until the user confirms it there.

UC-09: View Financial Health Score

Goal	See a single summary of how well money is being managed
Actor	User, with help from the Analytics Engine

Main flow	1. User selects Financial Health from the dashboard. 2. The system gathers the user's income, spending, budget adherence and saving rate for the recent period. 3. The system asks the Analytics Engine to calculate the health score. 4. The score is displayed along with the band it falls into, such as good, fair or poor. 5. The system shows a short explanation of the main factors behind the score. 6. The system displays how the score has moved compared with the previous period. 7. The system suggests one or two specific actions that would improve the score. 8. The user may open any factor to see the underlying figures. 9. The financial health score is displayed successfully.
Alternate Flow	Not Enough Data to Score At Step 3, if the account has too little history: 3a. System does not display a score. 3b. A message explains how much history is needed before a score can be shown. Score Has Dropped Sharply At Step 6, if the score has fallen by a large margin since the last period: 6a. System highlights the drop and names the factor most responsible for it. 6b. The user is offered a link to the budget page to review the limits.

UC-10: Export Financial Report

Goal	Download a report of income, expenses and budgets
Actor	User
Main flow	1. User selects Reports. 2. The system displays the export options. 3. The user chooses a date range and, optionally, specific categories. 4. The user chooses a format, either PDF or CSV. 5. The user selects Export. 6. The system checks that the selected range contains at least one record. 7. The system gathers the income, expense, budget and savings figures for that range. 8. The system generates the file with the totals matching what the dashboard displays. 9. The file is downloaded to the user's device. 10. A confirmation is displayed and the export is complete.

Alternate Flow**No Records in the Selected Range**

At Step 6, if the range contains no data:

- 6a. System does not generate a file.
- 6b. A message states that there is nothing to export for that range.
- 6c. User may pick a different range and the flow continues from Step 3.

Invalid Date Range

At Step 6, if the start date is after the end date:

- 6d. System refuses the range and asks the user to correct the dates.

Export Fails

At Step 8, if the file cannot be generated:

- 8a. System displays an error and does not download a partial file.
 - 8b. User may retry the export or choose the other format.
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